MILL TEST CERTIFICATE

AXIS
PIPE AND TUBE

XIS PIPE AND TUBE, LLC.
451 Louis E. Mikulin Rd.
ryan TX 77807

Customer: MRC Global PO Box Charleston,WV,25322 877-294-7574	**Sales Order:** 0630004951 **PO:** S 7AA 637379	**Invoice No.:** 1600049629 **Date of Issue:** 07/23/2024
Certificate No.:	1600049629-12	
Product Description:	5L 12.750 0.500 65.48# X52BDRH5BW-20P2	
Specification:	5L 46th Edition May 2019	

Heat Number	Lot Number	Mat Number	Steel Making / Coil Rolling
B402397	LT24185031	332758	Steel Dynamics Columbus, LLC , Columbus - MS - US / Steel Dynamics Columbus, LLC , Columbus - MS - US

Mechanical Properties

Tensile Test - Strip Specimen Gauge Length 2"

Loc. / Orien	Size	0.5%EUL Y.S(psi)	U.T.S. Body (psi)	U.T.S. Weld (psi)	Y/T Ratio	E.L. (%)	Hardness (HRB)
T180°	1 1/2"	63000	73900	76200	0.85	43.5	81

CVN — Temp (F) -20

	Body (ft-lbf)				Shear Area (%)			
	1	2	3	Avg	1	2	3	Avg
	215	211	224	217	100	100	100	100
Lateral Expasion (in)					UB	UB	UB	UB

CVN — Temp (F) -20

	Weld (ft-lbf)				Shear Area (%)				Size	Loc. / Orien	Acceptance Criteria (ft-lbf)(AVG)
	1	2	3	Avg	1	2	3	Avg			
	166	191	189	182	95	100	100	99	10x7.5 mm	3T90	15
Lateral Expasion (in)					0.089	UB	UB	0.089			

Chemical Composition (Max %)

Analysis (H/P)	C	Si	Mn	P	S	Ni	Cr	Mo	Cu	V	Al	Nb	Ti	B	Ca	N	CEpcm (API Max. A.C.)	CE Results
H	0.06	0.220	0.84	0.009	0.002	0.06	0.08	0.010	0.13	0.003	0.018	0.028	0.014	0.0001	0.0018	0.0070	0.25	0.12
P	0.07	0.215	0.77	0.011	0.002	0.06	0.08	0.010	0.13	0.003	0.011	0.023	0.015	0.0001	0.0030	0.0055	0.25	0.13
P	0.07	0.218	0.77	0.011	0.001	0.06	0.08	0.010	0.13	0.003	0.012	0.023	0.015	0.0002	0.0037	0.0054	0.25	0.13

Following Controls / Tests have been satisfactorily performed:
1.- Hydrostatic Testing: 4,090.00 psi Pressure for 5.00 seconds
2: Ultrasonic Tested to N10 - passed
3: Seam Annealed min 1,600 F
4: Flattening Test Passed

Remarks:
1: This material meets:Grades ASTM A53/A53M-22 Grade B
API 5L HFW PSL2 X52M,
2: This product meets the requirements of ASME BPVC.II.A Specification SA-53B-2023
3: Made and Melted in USA
4: Date of Manufacture 07/2024
5: This product is silicon and/or aluminum fully killed.
6: Certificate meets the requirements of ISO 10474:2013 (Inspection Certificate 3.1.B)
7: This material meets the requirements of NACE MR0175-2021 / ISO 15156:2020

Acronyms:
L = Longitudinal
Y.S. = Yield Strength
E.L. = Elongation
P = Product Analysis
UB = Un-Broken
T = Transverse
U.T.S. = Ultimate Tensile Strength
H = Heat Analysis
A.C. = Acceptance Criteria
N/A = Not Applicable

We hereby certify that the material herein has been made and tested in accordance with the above specification and also with the requirement called for by the above order.

Quality Manager